

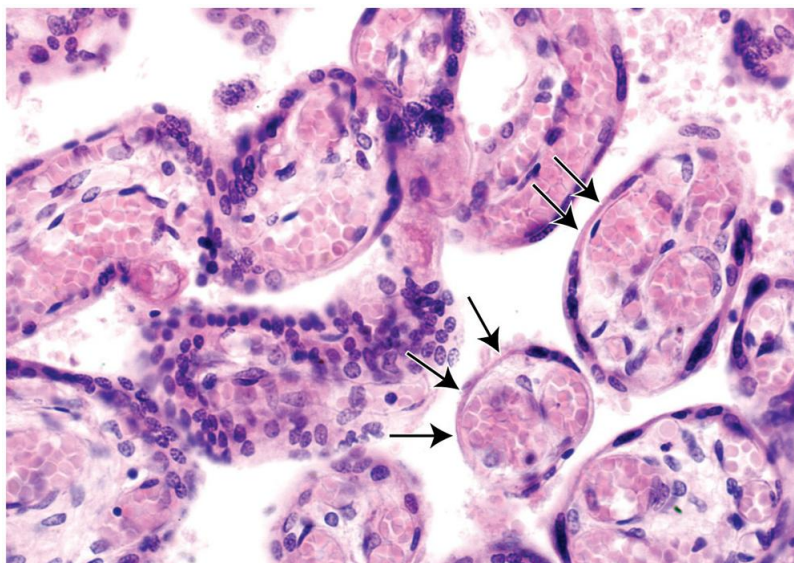
of Questions: 30

Question #: 1

An 18-year-old woman is brought to the emergency department because of a 2-day history of heavy vaginal bleeding. She is at 16 weeks' gestation and has a history of high blood pressure. Physical examination shows blood pooled in the posterior fornix of the vagina and an enlarged uterus. Laboratory studies show a positive serum beta human chorionic gonadotropin test (pregnancy test). An ultrasound shows a thickened endometrium with debris in the uterine cavity. A diagnosis of spontaneous abortion with retained fetal tissue is made. Surgical clearing of the endometrium is carried out to alleviate the bleeding. A photomicrograph of tissue obtained from uterus is shown. Which of the following hormones is secreted by the indicated structure?

- A. Melatonin
- B. Thyroxine
- C. Progesterone
- D. Androstenedione
- E. Follicle stimulating hormone

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Item Description: HCB.1167 - Placenta secretion

Question #: 2

A 19-year-old man is brought to the emergency department by the paramedics because of difficulty closing his jaw, for past 2 hours. Physical examination shows dislocated mandible. Radiological examinations shows mandibular head is displaced beyond the articular tubercle of the temporomandibular joint. Which of the following muscles most likely contracted as a result of this condition?

- A. Masseter
- B. Temporalis
- C. Medial pterygoid
- D. Lateral pterygoid

Item Description: Lateral pterygoid muscle

Question #: 3

A 21-year-old woman is brought to the emergency department following a car crash where she impacts the steering wheel. Physical examination shows abdominal bruising and guarding on palpation. Imaging studies show a lacerated pancreas with the digestive enzymes spilling anterior to it. Into which of the following locations will the contents of the lacerated organ most likely spill?

- A. Supracolic part of the greater sac
- B. Omental bursa
- C. Subhepatic bursa
- D. Infracolic part of the greater sac
- E. Subhepatic recess

Item Description: A 21-year-old woman

Question #: 4

A 1-month old boy is brought to the physician by his mother for a routine examination. She points out a small opening in the skin just in front of his left ear. The physician tells her not to be concerned because this is simply the result of a harmless condition in the development of the external ear. Which of the following is the most likely diagnosis?

- A. Pre-auricular sinus
- B. Keloid
- C. Auricular hillock
- D. Auricular cyst
- E. Auricular fistula

Item Description: A 1-month old boy is

Question #: 5

A 22-year-old man is brought to the emergency department because of severe bleeding from a head injury after a car crash. Physical examination shows the three fused layers of the scalp has been peeled back as a single layer. Which of the following layers are most likely involved?

- A. Skin, aponeurosis, pericranium
- B. Skin, loose connective tissue, pericranium
- C. Skin, loose connective tissue, aponeurosis
- D. Skin, dense connective tissue, aponeurosis
- E. Skin, loose connective tissue, dense connective tissue

Item Description: A 22-year-old man is

Question #: 6

An 18-year-old man is brought to the emergency department after a crash where his forehead absorbed most of the impact. Imaging studies show multiple cranial fractures. Physical examination shows a clear nasal discharge indicating a fracture through the anterior cranial fossa. Which of the following nerves is most likely

damaged?

- A. Optic
- B. Facial
- C. Olfactory
- D. Hypoglossal
- E. Maxillary

Item Description: An 18-year-old man i

Question #: 7

A 15-year-old boy is brought to the physician by his mother because he is unable to rotate his head to the left side. He suffered a stab wound to his neck just posterior to the sternocleidomastoid 3 weeks ago. Which of the following nerves was most likely damaged?

- A. Right spinal accessory
- B. Left spinal accessory
- C. Right vagus
- D. Left vagus
- E. Ansa cervicalis

Item Description: Spinal accessory nerve

Question #: 8

A 15-year-old girl comes to the physician because of a 24-hour history of headache, facial pain along her cheekbones and vision problems. She has an infected pimple on the ala of her nose which she has been repeatedly pick at. She is diagnosed with cavernous sinus thrombosis. Which of the following veins is the most likely route that introduced the infection to the cranial cavity?

- A. Angular vein

- B. Superior sagittal sinus
- C. Lingual vein
- D. Superior petrosal sinus
- E. Internal jugular veins

Item Description: Danger triangle of face

Question #: 9

A 25-year-old man comes to the physician because of numbness of the lower lip, left side of his face just anterior to his ear and difficulty chewing his food, for past 5 days. The nerve responsible for innervation to the affected region most likely passes through which of the following foramen?

- A. Foramen ovale
- B. Foramen rotundum
- C. Internal acoustic meatus
- D. Stylomastoid foramen
- E. Mandibular foramen

Item Description: Trigeminal nerve route

Question #: 10

A 50-year-old man comes to the emergency department because of a 2-week history of a pustular, draining abscess on his lower lip. He says he punctured the middle of his lip with a rusty nail and the wound is not closing spontaneously. Physical examination shows an open abscess. Which of the following lymph nodes is most likely to be enlarged on palpation?

- A. Inferior deep cervical
- B. Occipital
- C. Preauricular
- D. Submental

- E. Superior deep cervical
- F. Supraclavicular

Item Description: Neck lymph nodes

Question #: 11

A 50-year-old man comes to the physician because of a 1-week history of painful salivation. Physical examination shows a stone blocking the right submandibular duct and surgery is scheduled to remove the stone. During the surgery, there is trauma to the nerve that wraps around the submandibular duct. Which of the following symptoms will most likely be present in the patient as a result of this nerve damage?

- A. Weakened ability to protrude the tongue
- B. Weakened ability to depress the mandible
- C. Weakened ability to elevate the mandible
- D. Decreased general sensation from the anterior 2/3rds of the tongue on the affected side
- E. Decreased general sensation from the posterior 1/3rd of the tongue on the affected side

Item Description: Lingual nerve pathway

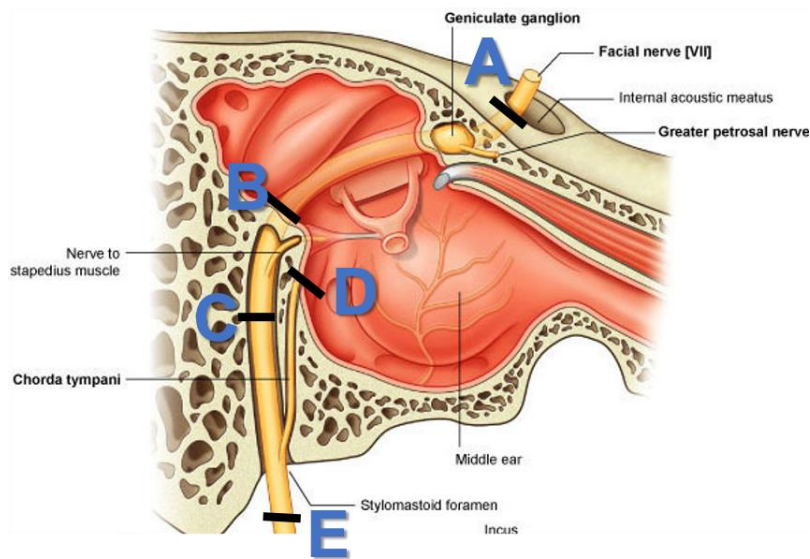
Question #: 12

A 40-year-old man come to the physician because of a 3-day history of loud sounds, impaired taste sensation and dry mouth. He was struck in the head by flying debris shortly before his current symptoms. Physical examination shows flattening of the nasolabial folds. In which of the following labeled areas (see image) is the lesion most likely located?

- A. A
- B. B

- C. C
- D. D
- E. E

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Item Description: Facial Nerve pathway through middle ear

Question #: 13

A 16-year-old boy is brought to the physician by his parents because of a 1-day history of fever, drowsiness and a confused mental state. He is admitted to hospital and diagnosed with a cavernous sinus infection and thrombosis resulting from an abcess in the infratemporal fossa. Which of the following is most likely the path taken by the infection from the fossa to the cavernous sinus?

- A. Ophthalmic vein
- B. Facial vein
- C. Pterygoid venous plexus
- D. Occipital emissary vein
- E. Maxillary vein

Item Description: A 16-year-old boy is

Question #: 14

A 44-year-old man comes to the physician because of a 1-week history of drooping of the right corner of his mouth. He can clench his teeth and chew on demand. Physical examination shows slight hearing loss and difficulty maintaining his balance while standing on one foot. His blink reflex is absent on the right, but sensation is normal on his entire face. Imaging studies show a tumor in the posterior cranial fossa. Which of the following additional symptoms is most likely present?

- A. Loss of touch sensation on the anterior two thirds of the tongue
- B. Mydriasis
- C. Loss of taste on the posterior third of the tongue
- D. Inability to shrug the shoulders
- E. Decreased ability to produce tears

Item Description: A 44-year-old man co

Question #: 15

A 22-year-old woman is brought to the emergency department because of a head wound after she fainted. Physical examination shows a 4cm deep laceration to the back of her head that is bleeding freely. Branches of which of the following arteries is most likely damaged?

- A. Occipital
- B. Deep temporal
- C. Posterior auricular
- D. Vertebral
- E. Superficial temporal

Item Description: A 22-year-old woman

Question #: 16

A 65-year-old man come to the physician because of a small lump on the right side of face, anterior to his right ear. At first the physician thought it might be just an enlarged lymph node. Physical examination shows no sign of infection around ear and scalp area, but the small lump is superficial to the parotid salivary gland, anterior to the external ear and posterior to the masseter muscle. Diagnosis of parotid gland cancer is made. Which of the following is the most likely risk of surgical removal of tumor?

- A. Lose of taste on the right side of tongue
- B. Lose of chewing on the right side
- C. Tongue deviating to the right side on protrusion
- D. Right side facial paralysis

Item Description: Parotid gland cancer

Question #: 17

A 4-year-old girl is brought to the physician by her parents because of a 3-month history of easy fatigability while playing. Her parents say that she has a habit of sucking on yellow paint chips she finds in their old barn. Physical examination shows conjunctival pallor. Which of the following enzymes is most likely inhibited in this child?

- A. Delta-aminolevulinic acid dehydratase
- B. Xanthine oxidase
- C. Glutathione peroxidase
- D. Delta-aminolevulinic acid synthase
- E. Dopamine beta-hydroxylase
- F. Homocysteine methyl transferase

Item Description: Heme biosynthesis

Question #: 18

A 20-year-old man comes to the physician because of a 1-month history of

seizures. He is prescribed valproic acid which acts by inhibiting the GABA-transaminase enzyme and increasing levels of GABA. Which of the following enzymes is most likely required in the presynaptic neurons for the synthesis of this neurotransmitter?

- A. Monoamine oxidase
- B. DOPA decarboxylase
- C. Glutamate decarboxylase
- D. Acetylcholinesterase
- E. Histidine decarboxylase
- F. Tryptophan hydroxylase

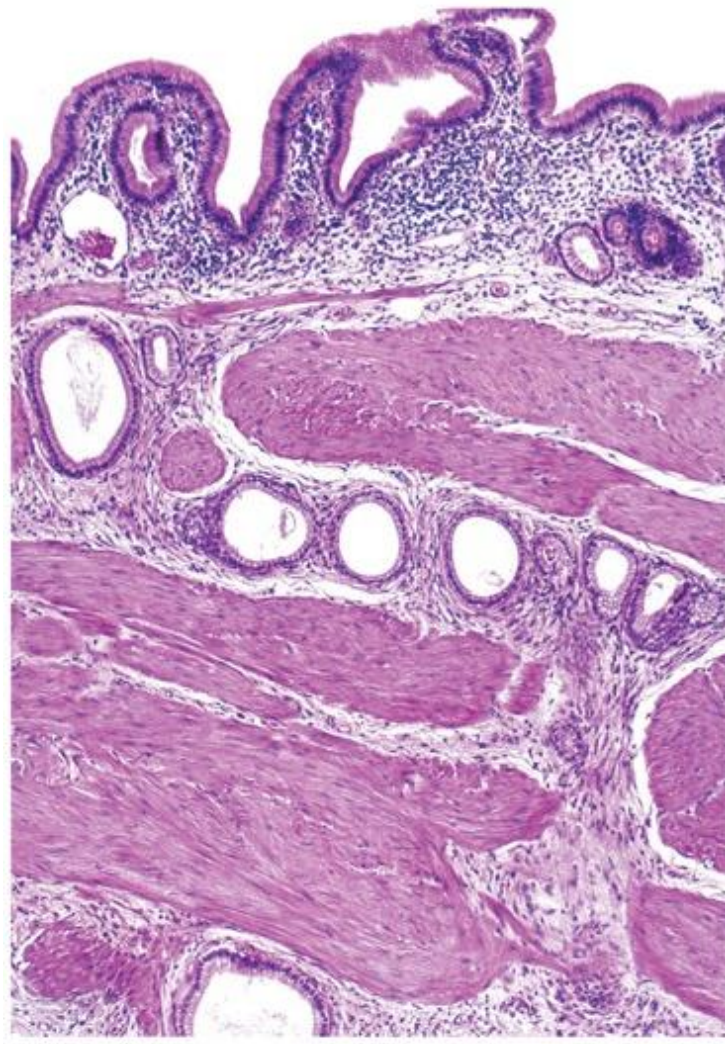
Item Description: Glutamate decarboxylase

Question #: 19

A 45-year-old woman comes to the physician because of a 2-month history of intermittent upper abdominal discomfort and bloating which occurs after meals. Physical examination of the abdomen shows mild tenderness in the right upper quadrant. Imaging studies show enlargement and thickening of the wall of a particular gastrointestinal organ. The organ is removed via laparoscopic surgery. A photomicrograph of the resected organ is shown. Which of the following gastrointestinal structures is most likely affected?

- A. Stomach
- B. Duodenum
- C. Jejunum
- D. Gallbladder
- E. Ileum
- F. Esophagus

Attachment:



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Item Description: HCB.1296 - Structure of the gall bladder

Question #: 20

A 24-year-old man comes to the physician because of a 3-month history of intermittent headaches, weight gain, and change in his shoe size. His blood pressure is 146/94 mmHg; other vital signs are within normal limits. Physical examination shows coarse skin, prominent nose, protruding lower jaw, and enlarged hands and feet. MRI of the brain shows an 8-mm mass in the Sella turcica. Laboratory studies show increased plasma IGF-1. Which of the following is most likely decreased in this patient?

A. Plasma growth hormone

- B. Plasma free fatty acids
- C. Synthesis of proteins in skeletal muscles
- D. Sensitivity to an oral glucose load
- E. Uptake of amino acids by skeletal muscle

Item Description: PHYS.1013. Glucose load in acromegaly

Question #: 21

An 80-year-old woman comes to the physician because of a 3-month history of fatigue and shortness of breath. Her vital signs are within normal limits. Physical examination shows a pale tongue and conjunctiva. Laboratory studies show low hemoglobin concentration, elevated mean corpuscular volume, and a low reticulocyte count. A peripheral blood smear shows macrocytic hypochromic red cells and hypersegmented neutrophils. Gastric juice studies show a decreased basal and maximal acid secretion. She is diagnosed with megaloblastic anemia due to intrinsic factor deficiency. Which of the following cells in the stomach is most likely defective in this patient?

- A. Chief cells
- B. G cells
- C. Goblet cells
- D. Mucous neck cells
- E. Parietal cells

Item Description: PHYS.1123. Intrinsic factor cellular source

Question #: 22

A 48-year-old woman is brought to the emergency department because of a 30-minute history of a deep penetrating wound to her right knee. Prior to surgery, a local anesthetic is administered via lumbar puncture, leading to analgesia and paralysis. Which of the following events is most likely directly affected by the drug?

- A. Resting membrane potential
- B. After-hyperpolarization

- C. Receptor potential
- D. Postsynaptic potential
- E. Action potential

Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0035 Action potential lidocaine

Question #: 23

A 45-year-old woman comes to the physician because of a 6-hour history of a traumatic injury to the back of her head. Over 24 hours, confusion and weakness arise. CT scan of the head shows a crescent-shaped hematoma that crosses suture lines, compressing the brain parenchyma. Between which of the following structures is the blood most likely accumulating in this patient?

- A. Dura mater and arachnoid
- B. Arachnoid and pia
- C. Inner and outer dura layers
- D. Pia mater and brain parenchyma
- E. Skull and dura mater

Item Description: SOM.MK.I.BPM2.2.NB.1.NSCI.0071 Intracranial hemorrhages

Question #: 24

A 46-year-old man comes to the physician because of a 2-day history of severe pain in his right knee. He says that his knee briefly "locked" when he got out of his car. He undergoes arthroscopic surgery for repair, requiring the application of a local anesthetic via lumbar puncture resulting in anesthesia and paralysis of his lower body. Which of the following structures is most likely targeted by the anesthetic?

- A. Nicotinic receptors
- B. Muscarinic receptors
- C. Voltage-gated sodium channels
- D. Ligand-gated calcium channels
- E. Voltage-gated calcium-channels

- F. Pressure-gated sodium channels
- G. Ligand-gated chloride channels
- H. Glycine receptors

Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0035 Explain the action-potential blocking effects of lidocaine.

Question #: 25

A group of researchers investigates relations among ionic conductance and membrane potential. A drug that blocks potassium channels is applied to a cultured neuron. Which of the following outcomes is most likely?

- A. Depolarization
- B. Hyperpolarization
- C. Increased conduction velocity
- D. Decreased conduction velocity
- E. Initiation of action potentials
- F. Prevention of action potentials

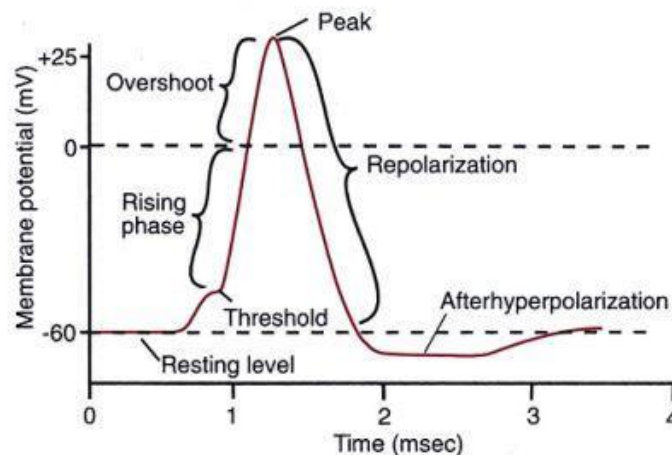
Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0027 Excitable cells signal conversion

Question #: 26

A researcher is studying the excitability of basal ganglia neurons. He finds that during a certain portion of the action potential it is possible to stimulate the neuron and produce another action potential but only if a greater-than-normal excitatory input is applied. Which of the following events most likely overlaps with the indicated period of reduced excitability (see figure)?

- A. Rest
- B. After-hyperpolarization
- C. Overshoot
- D. Repolarization
- E. Rising phase

F. Peak
Attachment:



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Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0038 Action potential conductances

Question #: 27

A 21-year-old woman comes to the physician because of a 6-hour history of painful cutaneous burns to her back caused by ultraviolet radiation. The sensation is of a throbbing quality. In such cases, propagation of the pain-eliciting signals from the skin to the spinal cord is slow but continuous. Which of the following pairs of properties most likely applies to the cells involved in the transmission of injury-related action potentials to the spinal cord?

- A. Pseudounipolar; myelinated
- B. Pseudounipolar; unmyelinated
- C. Multipolar; myelinated
- D. Multipolar; unmyelinated
- E. Bipolar, myelinated
- F. Bipolar, unmyelinated

Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0034 Conduction velocity myelination

Question #: 28

A 61-year-old man is brought to the emergency department by his son because of a 3-hour history of tingling, numbness and loss of control of his legs. Laboratory studies and nerve conduction test results are shown in the tables below. Which of the following is the most likely diagnosis?

Parameter	Measured value	Normal Range
K+	4.1 mM/L	3.5-5.0
Na+	143 mM/L	136-145
Cl-	98 mM/L	95-105

Test	Measured Value	Normal value
Peripheral nerve conduction velocity	20 m/s	40-60

- A. Hyperkalemia
- B. Hypernatremia
- C. Demyelination disorder
- D. Transient ischemic attack
- E. Traumatic injury

Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0034 Conduction velocity myelination

Question #: 29

A 63-year-old man comes to the physician because of a 1-year history of fatigue and pain. He says that over-the-counter preparations are ineffective in controlling pain. He receives a prescription for a drug that inhibits the uptake of serotonin. Which of the following nuclei is most likely integral to the synthesis of the targeted transmitter?

- A. Locus ceruleus
- B. Substantia nigra

- C. Nucleus basalis of Meynert
- D. Nucleus raphe magnus
- E. Nucleus accumbens

Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0107 Catecholamines and serotonin

Question #: 30

A 23-year-old woman comes to the physician because of a 5-year history of poorly controlled seizures. She is prescribed a new drug that acts as an agonist of an inhibitory amino acid. Which of the following neurotransmitters is most likely targeted by this new drug?

- A. Serotonin
- B. Dopamine
- C. GABA
- D. Glutamate
- E. Norepinephrine

Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0106 Amino acid transmitters